

2-6. Stability of hyperbolic toral automorphism.

semiconjugacy between $f: S^1 \rightarrow S^1$ ($\deg f = k$) & E_k

$\mathcal{C} = \{h: [0,1] \rightarrow [0,1] : \text{conti.}, h(0)=0, h(1)=1\}$

Fixed pt of a contraction on $\mathcal{C}$

i.e. $\exists \text{ surj. } h \text{ s.t.}$

Thm A: Any hyperbolic linear automorphism F_L of $\mathbb{T}^2$ is a factor of any homeo g in the same homotopy class

$F_L \circ h = h \circ g$

via a uniquely determined semiconjugacy h (homotopic to id)

g close to F_L in $C^0 \Rightarrow$ semiconjugacy close to id in C^0 .

Prop. Any C^1 map g suff. close to F_L in C^1 top.

is a factor of F_L .

$\Downarrow$

Thm B. $F_L: C^1$ strongly structurally stable.

i.e. $\mathcal{C}$ is structurally stable $\iff \exists$ nbd U of F_L in C^1 top.

s.t. $\forall g \in U: \text{top. conj. to } F_L$

& $\forall g \in U \Rightarrow \exists \text{ conj. homeo } h = h_g$.

s.t. h_g & $h_g^{-1} \rightarrow \text{id}$ (uniformly) if $g \rightarrow f$ (C^1)

Rmk. $\left\{ \begin{array}{l} \text{Thm A.} \Rightarrow [F_L]_{\text{homotopy}} : \text{semiconjugacy "to" } F_L \\ \text{Prop.} \Rightarrow C^1\text{-perturbation of } F_L : \text{semiconjugacy "from" } F_L \end{array} \right.$

$\Downarrow$ Thm B. "topological conjugacy".

NOT a direct result!

proof Thm A) lift of $F_L = L$. doubly periodic.
↓

$$g \in [F_L]_{\text{homotopy}} \Rightarrow \text{lift of } g = L + \tilde{g} \checkmark$$

$$h: \mathbb{T}^2 \rightarrow \mathbb{T}^2: \text{Cont.} \Rightarrow \text{lift of } h = \text{id} + \tilde{h}$$

If $h \circ g = F_L \circ h$, then

$$(\text{id} + \tilde{h}) \circ (L + \tilde{g}) = L \circ (\text{id} + \tilde{h}) \quad \text{Goal.}$$

$$\text{i.e. } \boxed{\text{id} + \tilde{h} = L^{-1} \circ (\text{id} + \tilde{h}) \circ (L + \tilde{g})}$$

$$L: \text{hyperbolic} \Rightarrow \lambda_u > 1, \quad 0 < \lambda_s < 1.$$

sol satisfies (6) in next page

$$\tilde{h} = h_1 v_u + h_2 v_s, \quad \tilde{g} = g_1 v_u + g_2 v_s. \quad (\because \tilde{g} \text{ given}).$$

$\mathcal{C}$: doubly periodic functions.

$$\mathcal{F}_1: \mathcal{C} \rightarrow \mathcal{C}, \quad \mathcal{F}_1 h = \lambda_u^{-1} g_1 + \lambda_u^{-1} h \circ (L + \tilde{g})$$

— fixed

$$\Rightarrow \|\mathcal{F}_1(h) - \mathcal{F}_1(h')\| = \lambda_u^{-1} \sup_{x \in \mathbb{T}^2} |h(L(x) + \tilde{g}(x)) - h'(L(x) + \tilde{g}(x))|$$

$$\leq \lambda_u^{-1} \sup_{y \in \mathbb{T}^2} |h(y) - h'(y)|$$

$$= \lambda_u^{-1} \|h - h'\|$$

$\Rightarrow \exists!$ fixed pt h_1 of $\mathcal{F}_1$.

$$\|h_1\| \leq \sum (\|\mathcal{F}_1^{n+1}(0) - \mathcal{F}_1^n(0)\|) = \frac{1}{1 - \frac{1}{\lambda_u}} \|\mathcal{F}_1(0)\| = \frac{\lambda_u}{\lambda_u - 1} \|g_1\|.$$

$$\mathcal{F}_2: \mathcal{C} \rightarrow \mathcal{C}, \quad \mathcal{F}_2 h = \lambda_s h \circ S - g_2 \circ S.$$

$$(S = (L + \tilde{g})^{-1})$$

$\Rightarrow \exists!$ fixed pt h_2 of $\mathcal{F}_2$.

$$\|h_2\| \leq \frac{1}{1-\lambda_s} \|g_2\|.$$

$$\begin{aligned} \begin{cases} h_1 = \lambda_u^{-1} g_1 + \lambda_u^{-1} h_1 \circ (L + \tilde{g}) \\ h_2 = \lambda_s h_2 \circ (L + \tilde{g})^{-1} - g_2 \circ (L + \tilde{g})^{-1} \end{cases} \quad (*) \\ \hookrightarrow h_2 \circ (L + \tilde{g}) = \lambda_s h_2 - g_2 \\ \Rightarrow h_2 = \lambda_s^{-1} g_2 + \lambda_s^{-1} h_2 \circ (L + \tilde{g}) \end{aligned}$$

$$\begin{aligned} \text{id} + \tilde{h} &= \text{id} + h_u v_u + h_s v_s \\ &= \text{id} + \lambda_u^{-1} g_1 v_u + \lambda_s^{-1} g_2 v_s \\ &\quad + \lambda_u^{-1} h_1 \circ (L + \tilde{g}) v_u + \lambda_s^{-1} h_2 \circ (L + \tilde{g}) v_s. \\ &= \underbrace{\text{id} + L^{-1} \cdot \tilde{g}}_{(= L^{-1} \circ \text{id} \circ (L + \tilde{g}))} + L^{-1} \cdot \tilde{h} \cdot (L + \tilde{g}) \\ &= L^{-1} \circ (\text{id} + \tilde{h}) \circ (L + \tilde{g}) \end{aligned}$$

• g : close to $\mathbb{F}_L$

$\Rightarrow \tilde{g}$: close to 0

$\Rightarrow \tilde{h}$: close to 0

$\Rightarrow h$: close to id .

proof of Prop.) Want: h s.t. $g \circ h = h \circ F_L$.

Goal.

$$\text{i.e. } (L + \tilde{g}) \circ (\text{id} + \mathcal{H}) = (\text{id} + \mathcal{H}) \circ L$$

$\mathcal{L}$: doubly periodic $\mathbb{R}^2 \rightarrow \mathbb{R}^n$

$$\mathcal{L}: \mathcal{E} \rightarrow \mathcal{E}, \quad \mathcal{L}(\mathcal{H}) := \mathcal{H} \circ L - L \circ \mathcal{H}$$

$$\begin{cases} \mathcal{L}_1(h_1) = h_1 \circ L - \lambda_1 h_1 \\ \mathcal{L}_2(h_2) = h_2 \circ L - \lambda_2 h_2. \end{cases}$$

$$\text{where } \mathcal{H} = h_1 v_1 + h_2 v_2.$$

$$\Rightarrow \mathcal{L}(\mathcal{H}) = \mathcal{L}_1(h_1) v_1 + \mathcal{L}_2(h_2) v_2.$$

$\exists$ inverse of $\mathcal{L}_1, \mathcal{L}_2$

$$\begin{aligned} \mathcal{L}_1^{-1}(h_1) &= -\sum \lambda_1^{-(n+1)} h_1 \circ L^n \\ \mathcal{L}_2^{-1}(h_2) &= \sum \lambda_2^n h_2 \circ L^{-(n+1)} \end{aligned} \quad \begin{pmatrix} \lambda_1 \geq 1 \\ 0 < \lambda_2 < 1 \end{pmatrix}$$

$$\text{Goal} \Leftrightarrow \mathcal{L}(\mathcal{H}) = \tilde{g} \circ (\text{id} + \mathcal{H}).$$

$$\text{invertible} \downarrow \quad \underbrace{\quad}_{\mathcal{T}(\mathcal{H})}$$

$$\Rightarrow \mathcal{H} = \mathcal{L}^{-1} \mathcal{T}(\mathcal{H}).$$

$$\|\mathcal{T}(h) - \mathcal{T}(h')\| = \sup_{x \in \mathbb{R}^2} |\tilde{g}(h(x)) - \tilde{g}(h'(x))|$$

$$\leq \|D\tilde{g}\| \sup_{x \in \mathbb{R}^2} |h(x) - h'(x)|$$

$$= \|D\tilde{g}\| \cdot \|h - h'\|.$$

$$\Rightarrow \|\mathcal{L}^{-1} \mathcal{T}\| \leq \|\mathcal{L}^{-1}\| \cdot \|D\tilde{g}\| \quad (\text{i.e. } \|D\tilde{g}\| \ll 1 \Rightarrow \exists \text{ fixed pt } \mathcal{H})$$

3.1. Growth of orbits.

$$P_n(f) := \# \text{ of } x \text{ s.t. } f^n(x) = x.$$

$$\rho(f) := \limsup_{n \rightarrow \infty} \frac{\log(\max(P_n(f), 1))}{n} : \quad \text{exponential growth rate for the seq. } P_n(f).$$

$$\rho(f) = 0 \Rightarrow \text{Consider "polynomial growth"}$$

$$\rho(f) < \infty \Rightarrow \zeta\text{-function of } f$$

$$\zeta_f(z) = \exp\left(\sum \frac{P_n(f)}{n} z^n\right) \quad z \in \mathbb{C}$$

$$|z| < \exp(-\rho(f)) \Rightarrow \text{analytic}$$

$$\exists \text{ singularity on } |z| = \exp(-\rho(f)).$$

$$\exists \text{ analytic continuation to a meromorphic function (in many cases)}$$

$$\text{e.g. } E_m \text{ for } m \geq 1$$

$$P_n(E_m) = m^n - 1.$$

$$\rightarrow \rho(E_m) = \log m$$

$$\zeta_{E_m}(z) = \exp\left(\sum \frac{m^n - 1}{n} z^n\right) = \frac{1-z}{1-mz}$$

$$\text{meromorphic on } \mathbb{C}, \text{ single pole at } \frac{1}{m} = \exp(-\rho(E_m)).$$

$$\cdot \mathcal{O}_N, \mathcal{O}_N^R; \quad P_n = N^n$$

$$\zeta_{\mathcal{O}_N}(z) = \zeta_{\mathcal{O}_N^R}(z) = \exp\left(\sum \frac{N^n}{n} z^n\right) = \frac{1}{1-Nz}$$

$$\cdot F_L(x, y) = (x+y, x+y). \quad P_n(F_L) = \lambda_u^n + \lambda_s^n - 2.$$

$$\zeta_{F_L}(z) = \frac{(1-z)^2}{\left(1 - \frac{3+\sqrt{5}}{2}z\right)\left(1 - \frac{3-\sqrt{5}}{2}z\right)}.$$

For a flow φ^t ,

$P_T(\varphi^t)$ = # all periodic orbits of period $\leq T$.

$$p(\varphi^t) := \limsup_{T \rightarrow \infty} \frac{\log(\max(P_T(\varphi^t), 1))}{T}.$$

$$J_{\varphi^t}(z) = \prod_{\gamma} (1 - \exp(-z \ell(\gamma)))^{-1}.$$

$\uparrow$ smallest pos. period of γ .

Prop. X, Y : cpt metric sp.

$\varphi^t: X \rightarrow X$
 $\gamma^t: Y \rightarrow Y$ $\left\{ \begin{array}{l} \text{cont. flows w/o fixed pts} \end{array} \right.$

If $\varphi^t \sim \gamma^t$ (orbit equiv.) & $p(\gamma^t) = 0$

$\Rightarrow p(\varphi^t) = 0.$ $\exists$ homeo h s.t. $\varphi^t = h^{-1} \circ \gamma^t \circ h$
is a time change of φ^t .

sketch) time change: $\varphi^t \leftrightarrow \chi^t = \gamma^{\alpha}$

α has max, say $\alpha \leq M$

$\Rightarrow \{ \text{periodic orbits of } \varphi^t \text{ w/ period } \leq T \}$

$\hookrightarrow \{ \text{periodic orbits of } \gamma^t \text{ w/ period } \leq 2MT \}$

$$P_T(\varphi^t) \leq P_{2MT}(\gamma^t) \Rightarrow p(\varphi^t) \leq 2M \cdot p(\gamma^t).$$

"topological entropy": the most important numerical invariant.
(w.r.t. orbit growth.)

$f: (X, d) \rightarrow (X, d)$: cont. on cpt X .

d_n^f : increasing seq. of metrics.

$$d_1^f = d, \quad d_n^f(x, y) = \max_{0 \leq i \leq n-1} d(f^i(x), f^i(y)).$$

measures distance between $I_x^n = \{x, \dots, f^{n-1}(x)\}$, I_y^n .

$S_d(f, \varepsilon, n)$ = minimal # of ε -balls covering X .

$\uparrow$ (n, ε) -spanning set

$$h_d(f, \varepsilon) := \limsup_{n \rightarrow \infty} \frac{1}{n} \log S_d(f, \varepsilon, n).$$

$$h_{\text{top}}(f) := \lim_{\varepsilon \rightarrow 0} h_d(f, \varepsilon) : \text{topological entropy}$$

Remk. $h_{\text{top}}(f)$ does not depend on a metric d .

$\Rightarrow$ it is an invariant of topological conjugacy.

Prop. g : factor of $f \Rightarrow h_{\text{top}}(g) \leq h_{\text{top}}(f)$.

$$\therefore S_{d_X}(f, \varepsilon, n) \geq S_{d_Y}(g, \varepsilon, n).$$

Prop. Λ : closed f -inv. set $\Rightarrow h_{\text{top}}(f|_{\Lambda}) \leq h_{\text{top}}(f)$

$$X = \Lambda_1 \cup \dots \cup \Lambda_m \Rightarrow h_{\text{top}}(f) = \max_{1 \leq i \leq m} h_{\text{top}}(f|_{\Lambda_i})$$

$$h_{\text{top}}(f^n) = |n| h_{\text{top}}(f)$$

$$h_{\text{top}}(f \times g) = h_{\text{top}}(f) + h_{\text{top}}(g).$$

$$\underbrace{D_d(f, \varepsilon, n)}_{\text{minimal \# of subsets whose diameter } \leq \varepsilon \text{ \& covering } X} \leq \sum D_d(f|_{\Lambda_i}, \varepsilon, n) \leq m \cdot D_d(f|_{\Lambda_j}, \varepsilon, n)$$

minimal # of subsets whose diameter $\leq \varepsilon$ & covering X .

$$D_d(f, 2\varepsilon, n) \leq S_d(f, \varepsilon, n) \leq D_d(f, \varepsilon, n)$$

$$\hookrightarrow \lim_{\varepsilon \rightarrow 0} \lim_{n \rightarrow \infty} \frac{1}{n} \log D_d(f, \varepsilon, n) = h_{\text{top}}(f)$$

$N_d(f, \varepsilon, n) :=$ maximal # of pts in X w/ pairwise d_n^f -distance $\geq \varepsilon$
 $\uparrow$ (n, ε) -separated set.

(n, ε) -separated $\Rightarrow (n, \varepsilon)$ -spanning

$$N_d(f, 2\varepsilon, n) \leq S_d(f, \varepsilon, n) \leq N_d(f, \varepsilon, n).$$

$$\hookrightarrow \lim_{\varepsilon \rightarrow 0} \lim_{n \rightarrow \infty} \frac{1}{n} \log N_d(f, \varepsilon, n) = h_{\text{top}}(f).$$

Rmk. $S_d(f, \varepsilon, n), D_d(f, \varepsilon, n), N_d(f, \varepsilon, n)$
 $\downarrow$
 $h_{\text{top}}(f)$

$\Phi = \{\varphi^t\}_{t \in \mathbb{R}}$: flow.

$$d_{\tau}^{\Phi}(x, y) := \max_{0 \leq t \leq \tau} d(\varphi^t(x), \varphi^t(y))$$

$\uparrow$
non-decreasing metric

$\therefore h_{\text{top}}(\Phi)$: defined by using d_{τ}^{Φ} .

Prop. $h_{\text{top}}(\Phi) = h_{\text{top}}(\varphi')$.

• smooth arc $\gamma \Rightarrow$ growth of length $f^n(\gamma)$

$\Downarrow$ generalization

• Volume growth of k -cell in $M \Rightarrow$ smooth conj. inv.
sup over k

Thm (Yomdin-Nash).

$f: C^\infty$ map on a cpt mfd M .

$\Rightarrow h_{\text{top}}(f) = \text{Volume growth of } f.$

• π : group, $T = \{\gamma_1, \dots, \gamma_s\}$: generating set for π .

$F: \pi \rightarrow \pi$: endomorphism

$\gamma \in \pi \Rightarrow \gamma = \gamma_1^{\bar{\epsilon}_1} \dots \gamma_s^{\bar{\epsilon}_s} \gamma_1^{\bar{\epsilon}_{s+1}} \dots \gamma_s^{\bar{\epsilon}_{2s}} \dots \gamma_s^{\bar{\epsilon}_{ks}}$

$L(\gamma, T) = \min \text{ of } \sum |\bar{\epsilon}_j| \text{ over } \forall \text{ representations}$

$L_n(F, T) = \max L(F^n \gamma_i, T).$

$\Rightarrow h_A(F) := \lim_{n \rightarrow \infty} \frac{1}{n} \log L_n(F, T) : \text{algebraic entropy.}$

$\uparrow$ indep. of T .

Prop. π : fin. gen, F : endomorphism

$\Rightarrow h_A(\gamma_0^{-1} F \gamma_0) = h_A(F) \quad \forall \gamma_0 \in \pi.$

M : cpt conn. mfd, f : Conti. on M , α : path $p \rightsquigarrow f(p)$.

$\Rightarrow f_*^\alpha$: endomorphism on $\pi = \pi_1(M, p)$

given by $[\gamma] \mapsto [\alpha^\vee \cdot f(\gamma) \cdot \alpha]$.

$\Rightarrow h_*(f) := h_A(f_*^\alpha)$: fundamental group entropy

$\uparrow$ indep. of p & α .

Rmk. $h_*(f) \leq h_{\text{top}}(f)$.

$f: M \rightarrow M \Rightarrow f_{*,i}: H_i(M; \mathbb{R}) \rightarrow H_i(M; \mathbb{R})$.

($i=1$)

• Hurewicz $\mathcal{H} \circ P: \pi_1 \rightarrow \pi_1 / [\pi_1, \pi_1] \rightarrow H_1$

• $P: H_1(M; \mathbb{Z}) \rightarrow H_1(M; \mathbb{R})$ (onto integer lattice).

$\mathcal{H}_p := P \circ \mathcal{H} \circ P: \pi_1(M, p) \rightarrow H_1(M; \mathbb{R})$.

α : path from p to $f(p)$

$\Rightarrow f_{*,1} \circ \mathcal{H}_p = \mathcal{H}_p \circ f_*^\alpha$

i.e. $f_{*,1}$ restricted to integer lattice: factor of f_*^α .

(T : generating set for $\pi_1(M, p)$)

Choose a norm on $H_1(M; \mathbb{R})$

$\Rightarrow \exists C > 0$ s.t.

$$\|\mathcal{H}_p(\gamma)\| \leq C L(\gamma, T) \quad \forall \gamma \in \pi_1(M, p).$$

$$\Rightarrow \log \rho(f_{*,1}) \leq h_*(f).$$

spectral radius $\max |\lambda|$.

$$\|f_{*,1}^n(h_p(\gamma_i))\| = \|h_p(f_*^n \gamma_i)\| \leq C \cdot L(f_*^n \gamma_i) \leq C \cdot L_n(f_*, T).$$

$$\downarrow$$

$$\lambda^n \cdot \|h_p(\gamma_i)\|.$$

$$(\tilde{c} = \dim M).$$

$$M: \text{orientable} \Rightarrow H_i(M; \mathbb{R}) = \mathbb{R}.$$

$f_{*,1}$: multiplication by $\deg f$.

$$\Rightarrow \log |\rho(f_{*,1})| = \log |\deg f| \leq h_{\text{top}}(f) \quad \forall C^1 \text{ map } f.$$

$\therefore$ smooth map f w/ $|\deg f| \geq 2$

has "complicated asymptotic behavior"

$\Phi = \varphi^t: X \rightarrow X$: flow, $p \in X$, $\{\gamma_y : \text{path from } p \text{ to } y\}$

$\ell(x, T)$: closed loops consisting of $\gamma_{\varphi^T x}^{-1} \cdot \{\varphi^t x\}_{t=0}^T \cdot \gamma_x$

$$\text{i.e. } \ell(x, T) = \left\{ \begin{array}{c} \gamma_{\varphi^T x}^{-1} \\ \text{---} \\ p \xrightarrow{\gamma_x} x \xrightarrow{\varphi^t x} \varphi^T x \end{array} \right\} \subset \pi_1(X, p)$$

$$\pi(\Phi, p, T, T) := \# \ell(x, T)$$

$$h_{\text{hom}}(\Phi) := \limsup_{T \rightarrow \infty} \frac{\log \pi(\Phi, p, T, T)}{T} : \text{homotopical entropy}.$$

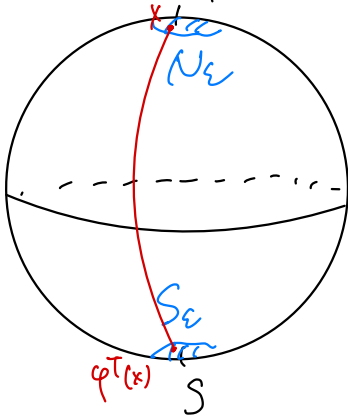
3.2. Examples of calculation of topological entropy.

* isometries

$$f: \text{isometry} \Rightarrow d_n^f = d \Rightarrow h_{\text{top}}(f) = 0.$$

Prop. translation, linear flow $\Rightarrow$ zero entropy.

* gradient flows. $\mathbb{F} = \{\varphi^t\}$



$$K := S^2 \setminus (N_\epsilon \cup S_\epsilon) : \text{cpt.}$$

$T(\epsilon) :=$ "longest" time from ∂N_ϵ to ∂S_ϵ .

$\left\{ \epsilon\text{-balls of orbit segment of time } T(\epsilon) \right\}$
starting at a pt on ∂N_ϵ

Covers $K \Rightarrow \exists$ finite subcover

$$\left\{ \varphi^t(y_1) \right\}_{t=0}^{T(\epsilon)}, \dots, \left\{ \varphi^t(y_{M(\epsilon)}) \right\}_{t=0}^{T(\epsilon)} \quad \leftarrow M(\epsilon)$$

$$\forall \epsilon > 0, \exists \delta(\epsilon) \text{ s.t. } d_T^{\mathbb{F}}(x, \varphi^s(x)) < \epsilon \quad \forall 0 \leq s \leq \delta(\epsilon).$$

$$T \gg 0 \Rightarrow \exists k \ (k \cdot \delta(\epsilon) > T \gg 0)$$

s.t. k times time $\delta(\epsilon)$ shifts of
are contained in N_ϵ or S_ϵ .

$$\text{minimal \# of } \epsilon\text{-balls of } d_T^{\mathbb{F}} \leq 3k \cdot M(\epsilon) \leq C \cdot T^{M(\epsilon)}$$

$$\therefore \left(d_T^{\mathbb{F}}(x, \varphi^{l \cdot \delta(\epsilon)}(y_i)) = \max_{0 \leq t \leq T} d(\varphi^t(x), \varphi^t(\varphi^{l \cdot \delta(\epsilon)}(y_i))) < \epsilon \right)$$

for some $i = 1, \dots, M(\epsilon), \quad l = -k, \dots, k$

Prop. gradient flow of round sphere $\Rightarrow$ zero entropy.

* expanding maps.

Prop. $|m| \geq 2 \Rightarrow h_{\text{top}} E_m = \log |m| = p(E_m)$.

$$\Sigma = m^{-k} \{ i m^{-n-k} : i = 0, \dots, m^{n+k} - 1 \}$$

↖ maximal (n, m^k) -separated set.

i.e. $N_d(E_m, m^k, n) = m^{n+k}$.

Cor. $f: S' \rightarrow S'$: expanding map of degree m

$$\Rightarrow h_{\text{top}}(f) = p(f) = h_{\text{inf}}(f) = \log |m|.$$

* Shifts & topological Markov chains.

$$(\Sigma_N, d_{\text{DN}}) \quad d_{\text{DN}}(w, w') = \sum \frac{|w_n - w'_n|}{(|0N|)^{|n|}}$$

$$\alpha = (\alpha_{-k}, \dots, \alpha_k)$$

$$C_\alpha^k = \{ w \in \Sigma_N : w_i = \alpha_i \} = \varepsilon_k = (|0N|)^{-k}/2 \text{-ball of any } w \in C_\alpha^k.$$

$$C = \underbrace{C_{\alpha_{-k}, \dots, \alpha_k, \dots, \alpha_{k+n}}}_{-k \sim k+n} : \varepsilon_k \text{-ball of any } w \in C$$

w.r.t. d_N^{DN} metric

$\Rightarrow$ # of possible words = N^{2k+n+1} - many distinct ε_k -ball.

$$\therefore S_{d_{\text{DN}}}(\Sigma_N, \varepsilon_k, n) = N^{2k+n+1}.$$

$$\Rightarrow h_{\text{top}}(\Sigma_N) = \lim_{k \rightarrow \infty} \lim_{n \rightarrow \infty} \frac{\log N^{2k+n+1}}{n} = \log N.$$

Prop. $h_{\text{top}}(\sigma_A) = p(\sigma_A) = \log \lambda_{\max}$ for any transitive TMC σ_A .

* hyperbolic toral automorphism.

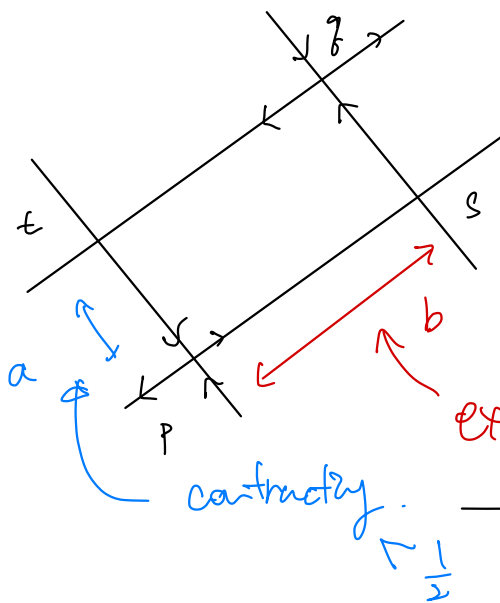
$$F_L(x,y) = (2x+y, x-y) \pmod{1}$$

$$A = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{pmatrix}$$

$$\lambda = \frac{3 \pm \sqrt{5}}{2}, \quad 3 \cdot 0.$$

F_L factors σ_A ($F_L \circ h = h \circ \sigma_A$)

$$\Rightarrow h_{\text{top}}(F_L) \leq h_{\text{top}}(\sigma_A) = \log \frac{3+\sqrt{5}}{2}$$



p, q : periodic pts (period n)

minimal rectangle

$\Leftarrow$ If $\text{dist}(p, q) < \frac{1}{4}$

expanding $\nearrow^2$

by F_L action.

$$\begin{aligned} a+b &\xrightarrow{F_L} \frac{a}{\lambda} + \lambda b \\ &\xleftarrow{F_L^{-1}} \lambda a + \frac{b}{\lambda} \end{aligned}$$

> at least one $> a+b$.

$\dots \rightsquigarrow$ rectangle becomes no longer minimal !!

$\therefore p, q$: periodic

$\Rightarrow$ n -periodic orbits form $(n, \frac{1}{4})$ -separated set

$$\therefore N_d(F_L, \frac{1}{4}, n) \geq P_n(F_L)$$

* finiteness of entropy of Lipschitz maps.

$$h_{\text{top}}(f) \leq D(X) \cdot \max(0, \log L(f))$$

↑ Lipschitz constant.

$f: M \rightarrow M: C^1$ on cpt Riemannian mfd

$$\Rightarrow h_{\text{top}}(f) \leq \max(0, \dim M \cdot \log \|Df\|) < \infty.$$

3.3. Recurrence properties.

Def. cpt $A \subset X$: attractor for f if $\exists$ nbd V of A & $N \in \mathbb{N}$
 s.t. $f^N(V) \subset V$ & $A = \bigcap f^n(V)$.

$y \in X$: ω -limit pt (resp. α -limit pt) for $x \in X$

if $\exists$ seq. of moments of time going to ∞ (resp. $-\infty$)
 s.t. images of x converge to y .

$$\omega(x) = \bigcap_{t \geq 0} \overline{\bigcup_{t \geq t} \varphi^t(x)}, \quad \alpha(x) = \bigcap_{t \leq 0} \overline{\bigcup_{t \leq t} \varphi^t(x)}$$

e.g. $f: S^1 \rightarrow S^1, \quad x \mapsto x + \frac{1}{t_0} \sin^2 \pi x \pmod{1}$

$x=0$: Not attractor, but $\alpha(0) = \omega(0) = \{0\}$.

Def. $x \in X$: positively recurrent (resp. negatively recurrent) if $x \in \omega(x)$ (resp. $x \in \alpha(x)$)
recurrent if $x \in \alpha(x) \cap \omega(x)$.

$R^+(t), \overset{\text{blue arrow}}{R^-(t)}, R(t)$: closure of pos./neg./- recurrent pts.

Def. $x \in X$: nonwandering w.r.t. $f: X \rightarrow X$ if $\forall$ open $U \ni x$,
 $\downarrow$ $NW(f)$ $\exists N > 0$ s.t. $f^N(U) \cap U \neq \emptyset$.

Prmk. $\exists N$ s.t. $n \geq N \Rightarrow f^n(U) \cap U = \emptyset \quad \forall$ open $U \ni x$

$\Rightarrow x$: non-wandering ($\because x$ is not a periodic pt.)

Prop. $NW(f)$: closed & f -inv.

$\nwarrow$ contains $\forall$ ω -, α -limit pts for all pts.

Cor. $f: X \rightarrow X \Rightarrow NW(f) \neq \emptyset$
 $\nwarrow$ cpt.

$$NW_1(f) = NW(f), \dots, NW_{n+1}(f) = NW(f|_{NW_n(f)})$$

$$NW_\omega(f) = \bigcap NW_n(f)$$

↓ again ... $\Rightarrow NW_{\omega+1}(f) = NW(f|_{NW_\omega(f)})$

↳ transfinite induction stops at a countable ordinal.

↖ "center" of the dynamical system.
 $NW(f)$ or $NW_2(f)$ in many cases.

$NW(f)$: "hub" of recurrence behavior.

↑ α - / ω - limits, recurrence pts, periodic pts.

$$h_{top}(f) = h_{top}(f|_{NW(f)}).$$

e.g. $f_\lambda: \mathbb{C} \rightarrow \mathbb{C}, z \mapsto (1-\lambda)z^2 \quad (\lambda \in [0,1])$

• $\lambda = 0$; $NW(f_0) = S^1 \cup \{0\} \leadsto h_{top} = \log 2$

• $\lambda > 0$; $NW(f_\lambda) = \{0\} \leadsto h_{top} = 0.$

Prop. conti. f on cpt metric sp. X has an inv. minimal set.

Cor. $R(f) \neq \emptyset.$

$$Per(f) \subset M(f) \subset R(f) \subset R^+(f) \cup R^-(f) \subset NW(f).$$

⌞
 closure of union of inv. minimal sets.

nontrivial recurrence : $\exists$ nonperiodic recurrent pts.

↳ low dim dynamics $\nwarrow$ homeo of circle
 flows on surfaces